STEP 1
Start with the time domain signal called $x(t)$

STEP 2
Multiply the time domain signal by an infinite number of exponential curves, each with a different decay constant, σ. That is, calculate the signal: $x(t)\, e^{-\sigma t}$ for each value of σ from negative to positive infinity.

STEP 3
Take the complex Fourier Transform of each exponentially weighted time domain signal. That is, calculate:

$$\int_{-\infty}^{\infty} [\, x(t)\, e^{-\sigma t}\,]\, e^{-j\omega t}\, dt$$

for each value of σ from negative to positive infinity.

STEP 4
Arrange each spectrum along a vertical line in the s-plane. The positive frequencies are in the upper half of the s-plane while the negative frequencies are in the lower half.

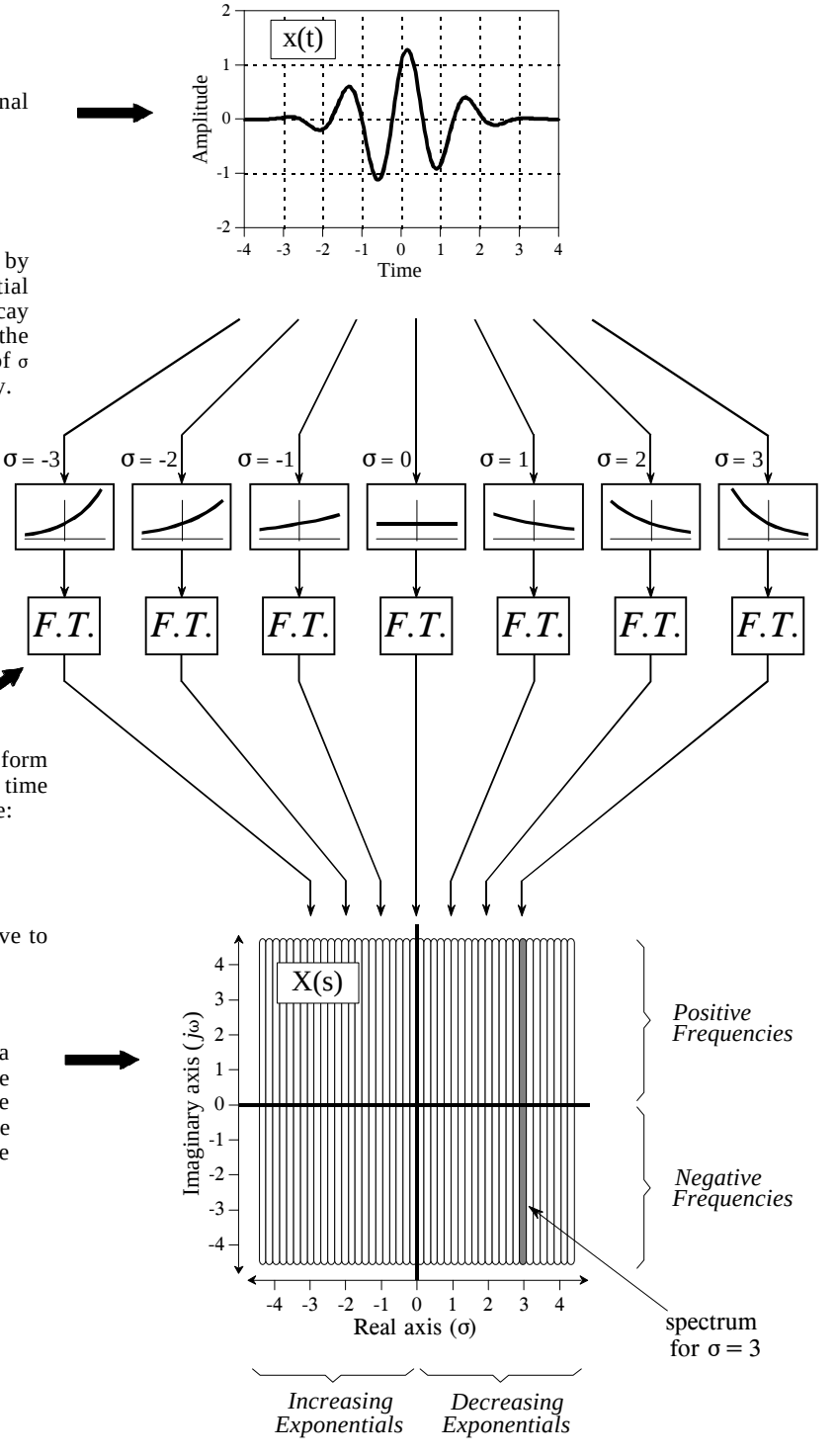

FIGURE 32-1
The Laplace transform. The Laplace transform converts a signal in the *time domain*, $x(t)$, into a signal in the *s-domain*, $X(s)$ or $X(\sigma,\omega)$. The values along each vertical line in the s-domain can be found by multiplying the time domain signal by an exponential curve with a decay constant σ, and taking the complex Fourier transform. When the time domain is entirely real, the upper half of the s-plane is a mirror image of the lower half.